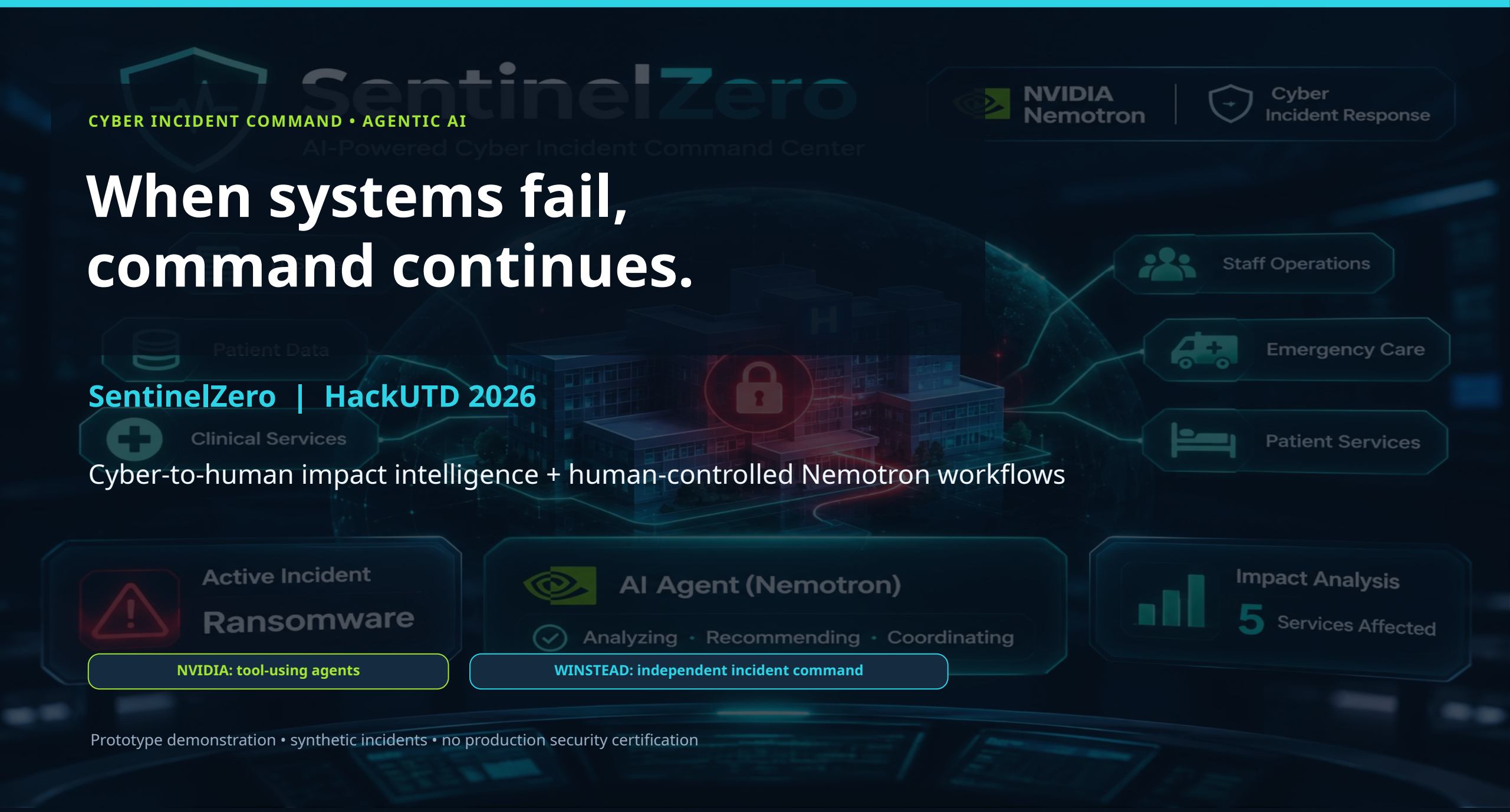
CYBER INCIDENT COMMAND • AGENTIC AI

When systems fail, command continues.

SentinelZero | HackUTD 2026

Cyber-to-human impact intelligence + human-controlled Nemotron workflows



NVIDIA: tool-using agents

WINSTEAD: independent incident command

Prototype demonstration • synthetic incidents • no production security certification

The attack is only half the emergency.

What happens after a cyber incident is detected determines how much disruption follows.

$4.99M

AVERAGE BREACH COST

IBM 2026 global study; market context

56%

INCREASE IN AI-ENABLED ATTACKS

IBM 2026; not a SentinelZero result

1 in 4

MALICIOUS BREACHES AI-ENABLED

IBM 2026 findings

The operational failure: the team loses a shared picture.

Compromised email • Fragmented evidence • Unclear service impact • Uncoordinated approvals

Sources: IBM 2026 Cost of a Data Breach Report, Jul 29 2026 (ibm.com / newsroom.ibm.com).

WHY NOW

Your normal tools may be compromised.

Winstead’s threat assumption: act as though an adversary can access internal systems.

EMAIL & IDENTITY

The attacker may see messages or disable sign-in.

INCIDENT RECORDS

Findings scatter across chats, spreadsheets, and tickets.

SERVICE CONTINUITY

A broken server can interrupt critical human workflows.

DECISION AUTHORITY

AI or responders must not execute actions without approval.

SentinelZero is designed as an independent response plane—not a replacement for SIEM, EDR, or forensic tools.

OUR ANSWER

One command picture. One accountable workflow.

SentinelZero bridges technical findings, operational continuity, human decisions, and evidence.

01	**DOCUMENT**	Structured incident states and scope
02	**UNDERSTAND**	Cyber → service → human impact
03	**COORDINATE**	Tasks, roles, secure team messages
04	**RECOMMEND**	Nemotron-powered multi-step analysis
05	**AUTHORIZE**	Commander approves; AI cannot self-execute

Eight surfaces. One incident response experience.

The codebase includes these dedicated interfaces; some currently use exercise-mode data.

COMMAND CENTER

KPIs + incident lifecycle

DATA SCOPE LEDGER

Locked vs exfiltrated data

IMPACT GRAPH

Interactive 2D + 3D map

TASKS & APPROVALS

Assigned actions + sign-offs

OUT-OF-BAND COMMS

Incident-specific team messaging

TEAM & ACCESS

Members + role permissions

AUDIT & EVIDENCE

Hash chain + evidence records

NEMOTRON CONSOLE

Four-stage AI investigation

Source: current main branch, apps/web/src/components and apps/web/src/pages (Nov 8, 2026).

OUR DIFFERENTIATOR

Cyber-to-Human Impact Graph

Not just “which server is down?” — “which essential service depends on it?”



Two graph modes, one evidence model

React Flow: accessible 2D / React Three Fiber: exploratory 3D / UNKNOWN ≠ SUSPECTED ≠ CONFIRMED

Illustrative graph relationships; fictional hospital exercise. Dependency does not prove patient harm.

Beyond chat: four focused agent stages

An orchestrator carries one shared execution budget and restricts tools by stage.

01

TRIAGE

Find verified facts and open questions

02

IMPACT

Trace assets → dependencies → services

03

CONTAINMENT

Propose next tasks, never execute

04

COMMS

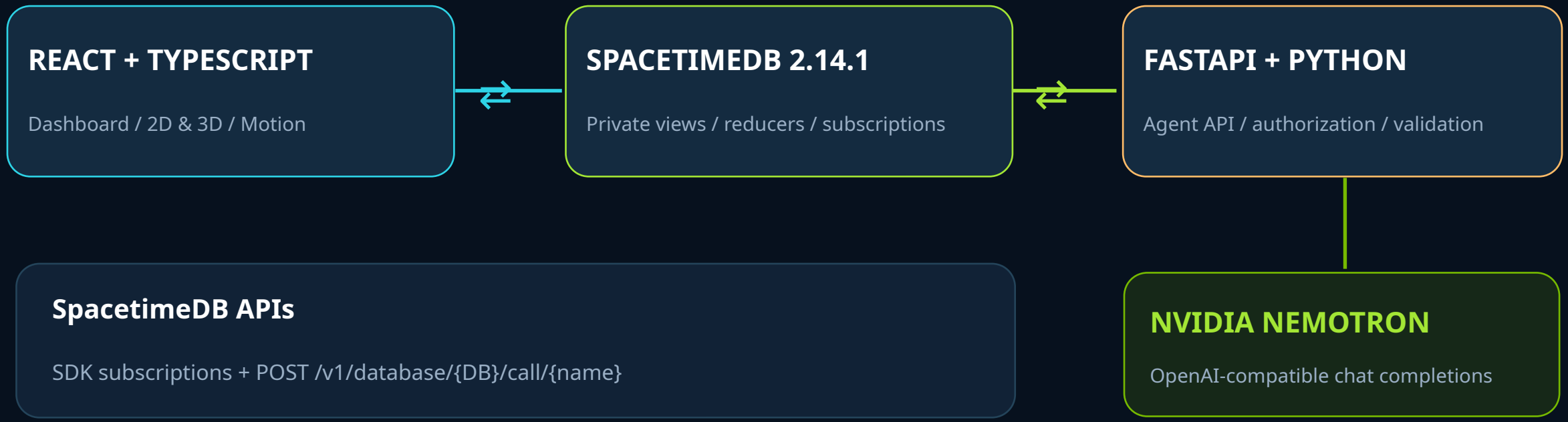
Draft an operational situation report

CONTROL PLANE: READ-ONLY TOOLS → VALIDATED OUTPUT → HUMAN APPROVAL

Source: services/ai/app/agents/orchestrator.py and subagents.py; default AI_AGENT_MODE=orchestrated.

Real APIs, explicit trust boundaries

Why each part exists—and what it is allowed to do.



API routes in docs/INTEGRATION_CONTRACT.md and services/ai/app/routes/analyze.py.

Designed for control when internal systems cannot be trusted.

Five required capabilities; a secure deployment must be independently verified.

1 ISOLATION	Separate host, credentials and identity plane	Design in code; deployed independence not verified
2 ACCESS CONTROL	Owner-managed members + five-role RBAC	Server-side permission tests recorded
3 INCIDENT DOCUMENTATION	Lifecycle, timeline, evidence, real-time state	Backend reducers and views present
4 COMPROMISED DATA SCOPE	Encryption vs exfiltration; UNKNOWN supported	Data Scope Ledger in UI + backend model
5 REAL-TIME COMMS	Incident chat and authorized message views	Backend implemented; live browser test needed

PROOF, NOT PROMISES

Engineering evidence: 202 recorded passing tests.

Local results reported in repository documentation; not a passing CI badge.

56
FRONTEND TESTS
Recorded in repo docs

27
BACKEND UNIT
Recorded in repo docs

40
BACKEND INTEGRATION
Recorded in repo docs

79
PYTHON AI TESTS
Recorded in repo docs

What we will not claim

Live NVIDIA inference, a fully verified browser-to-backend demo, production readiness or HIPAA compliance.

Source: README.md and docs/testing.md, Nov 8 2026. Live model test skipped; latest GitHub Actions run failed.

A hospital ransomware crisis—without a real victim.

Scenario data is entirely synthetic, repeatable, and explicitly labeled EXERCISE.

12
CYBER ASSETS
Fictional exercise

3
DISRUPTED SERVICES
Fictional exercise

4
HUMAN TEAMS
Fictional exercise

142k
SAMPLE ENCRYPTED PHI RECORDS
Fictional exercise

CRITICAL DISTINCTION

Encrypted ≠ exfiltrated. The scenario keeps exfiltration UNKNOWN until supported by evidence.

Sources: README.md Hope Valley fixtures; simulator/scenarios/hospital_ransomware.json (Riverbend scenario differs).

LIVE DEMONSTRATION

Three minutes. One story. No disconnected screens.

Follow the same fictional incident from detection through decision and audit.

0:00–0:30	**Command Center**	Open incident; identify priority and current state.
0:30–1:00	**Impact Graph**	2D/3D: asset → service → affected operational unit.
1:00–1:55	**Nemotron Pipeline**	Show stages, permitted tool calls, and evidence.
1:55–2:25	**Commander Approval**	Review proposal; approve once; verify task.
2:25–3:00	**Scope + Chat + Audit**	Show uncertainty, coordination, traceability.

Yes, there are established alternatives.

We compete on a different product thesis—not by pretending security orchestration is new.

Cortex XSOAR	Broad SOAR automations, war room, many integrations	Strong established enterprise platform
ServiceNow SIR	Security-to-IT workflows, case management, AI	Deep enterprise workflow ecosystem
TheHive	Collaborative incident case management	Established response case management
SentinelZero	Out-of-band command vision + cyber-to-human impact graph + proposal-only Nemotron	Focused prototype; narrower integrations

Sources: Palo Alto Networks Cortex XSOAR; ServiceNow Security Incident Response; TheHive/StrangeBee product pages.

WHY CHOOSE US

Built for the moment everything else is uncertain.

A focused command layer for response teams who need clarity more than another alert queue.

OPERATIONS FIRST

Trace technical compromise to the services and people who depend on them.

CONTROL FIRST

Only authenticated humans authorize consequential actions.

TRANSPARENCY FIRST

Separate evidence-backed facts from suspected impact and unknown exfiltration.

CONTINUITY FIRST

Designed to function outside compromised corporate infrastructure.

The product is for the responders—not for the demo.

What each stakeholder receives when the normal operating picture breaks down.

INCIDENT COMMANDER	See priorities, approve actions, hold teams accountable.
SECURITY ANALYST	Link evidence, uncertainty, assets, and recommended work.
OPERATIONS LEADER	Understand essential service impact without reading raw alerts.
RESPONSE TEAM	Receive clear assignments, communicate, and preserve a timeline.

THE DECISION

When the network is under attack, who is running the response?

SentinelZero is designed to be the answer.

One secure command view. One evidence trail. One human-owned decision process.

SEE THE LIVE DEMO • REVIEW THE SOURCE • CHALLENGE THE SECURITY MODEL

github.com/Saumya-Patel-10/SentinelZero

What is verified, simulated, and still unfinished.

Use this slide if judges ask about deployment, inference or reliability.

Backend auth, tests	Implemented + recorded integration tests
Four-stage agent	Implemented + scripted evaluation tests
Real Nemotron inference	Not run in repo evaluation (missing NVIDIA_API_KEY)
Frontend SDK end-to-end	Not demonstrated in repo testing docs; exercise fallbacks exist
Independent deployed isolation	Design requirement; no production deployment/pentest
CI on latest main	Latest GitHub Actions workflow failed
Customer outcome metrics	No measured time-savings, breach-reduction, or ROI

Reproducible references

Project claims come from source code; market and breach figures from cited vendors and IBM.

GitHub main	github.com/Saumya-Patel-10/SentinelZero
Integration / security	docs/INTEGRATION_CONTRACT.md, docs/SECURITY_MODEL.md
AI tests / limitations	docs/testing.md, docs/limitations.md, docs/AI_EVALUATION.md
IBM 2026 Cost of Data Breach	newsroom.ibm.com/2026-07-29-ibm-study-one-in-four-malicious-breaches-are-ai-enabled
NVIDIA Nemotron 3 Super	docs.api.nvidia.com/nim/reference/nvidia-nemotron-3-super-120b-a12b
Cortex XSOAR	paloaltonetworks.com/cortex/cortex-xsoar
ServiceNow Security Incident Response	servicenow.com/products/security-incident-response.html
TheHive	github.com/TheHive-Project/TheHive